

Computer Graphics (CSC214)

► Unit - 04: Three Dimensional Geometric Transformations

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Credit Hours: 3hrs

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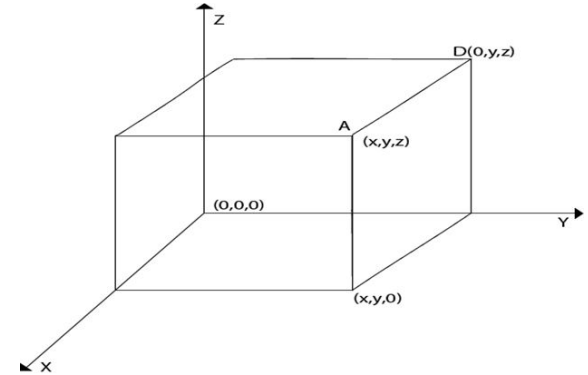


▶ 3D Transformation 4.1

► 3D Transformation:

3D computer graphics or three dimensional computer graphics are graphics that use a **three dimensional representation** of geometric data that is stored in the computer for the purpose of **performing calculations and rendering 2D images**.

- 2D is “flat” using the **horizontal and vertical (X & Y) dimensions**, the image has only two dimensions. 3D **adds the depth (Z)** dimension. This third dimension allows for rotation and visualization from multiple perspectives.
- We can perform different transformation by specifying the three dimensional transformation vector, however the 3D transformation is **more complex** than 2D transformation.



► 3D Transformation:

Q. What are the issue in 3D that makes it more complex than 2D?

- When we model and display a three-dimensional scene, there are many more considerations we must take into account besides just including coordinate value as 2D, some of them are:
 - Relatively **more coordinates point** are necessary.
 - Object boundaries can be constructed with **various combinations** of plane and curved surfaces.
 - Consideration of **projection** (dimension change with distance) and transparency.
 - Many consideration on visible surface detection and remove the hidden surfaces.

► 3D Transformation:

❖ **Homogeneous coordinates:** (x, y, z, w)

➤ 4 × 4 transformation matrices

$$\begin{bmatrix} x' \\ y' \\ z' \\ w' \end{bmatrix} = \begin{bmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix}$$

► Identity Matrix:

$$\begin{bmatrix} x' \\ y' \\ z' \\ w \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix}$$

► 3D Translation:

▪ 3D Translation

A point is translated from position $P(x, y, z)$ to position $P'(x', y', z')$ with the matrix operation as:

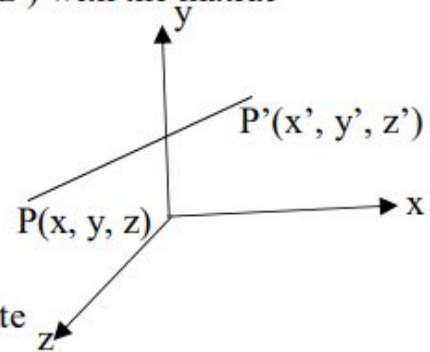
$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Parameter t_x, t_y, t_z specify transition distances for the coordinate directions x, y, z . This matrix representation is equivalent to three equations:

$$x' = x + t_x$$

$$y' = y + t_y$$

$$z' = z + t_z$$



► 3D Rotation:

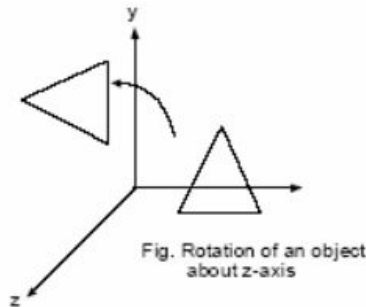
Rotation about Z-axis

$$x' = x \cos \theta - y \sin \theta$$

$$y' = x \sin \theta + y \cos \theta$$

$$z' = z$$

$$R_z = \begin{bmatrix} \cos \theta & -\sin \theta & 0 & 0 \\ \sin \theta & \cos \theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



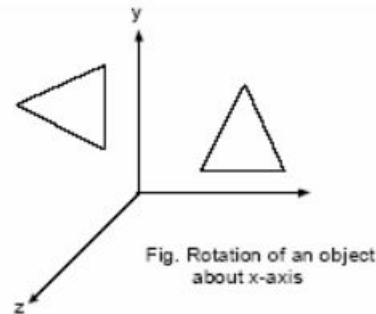
Rotation about x-axis

$$x' = x$$

$$y' = y \cos \theta - z \sin \theta$$

$$z' = y \sin \theta + z \cos \theta$$

$$\therefore R_x = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta & 0 \\ 0 & \sin \theta & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



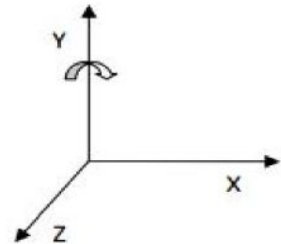
Rotation about y-axis

$$y' = y$$

$$z' = z \cos \theta - x \sin \theta$$

$$x' = z \sin \theta + x \cos \theta$$

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \theta & 0 & \sin \theta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin \theta & 0 & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$



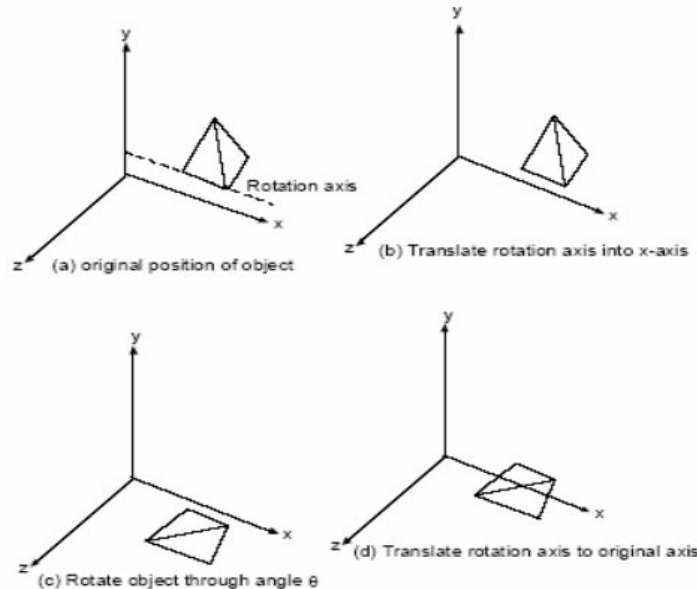
► General 3D rotation (Rotation about any arbitrary axis)

➤ Parallel to any of the co-axis:

When an object is to be rotated about an axis that is parallel to one of the co-ordinate axis, we need to perform some series of transformation.

1. Translate the object so that the rotation axis coincides with the parallel co-ordinate axis. $T(-a, -b, -c)$ where, (a, b, c) is any point on the rotation axis.
2. Performed the specified rotation about the axis. $R_x(\theta)$
3. Translate the object so that the rotation axis is moved to its original position. $T(a, b, c)$.

Net transformation = $T(a, b, c) R_x(\theta) T(-a, -b, -c)$



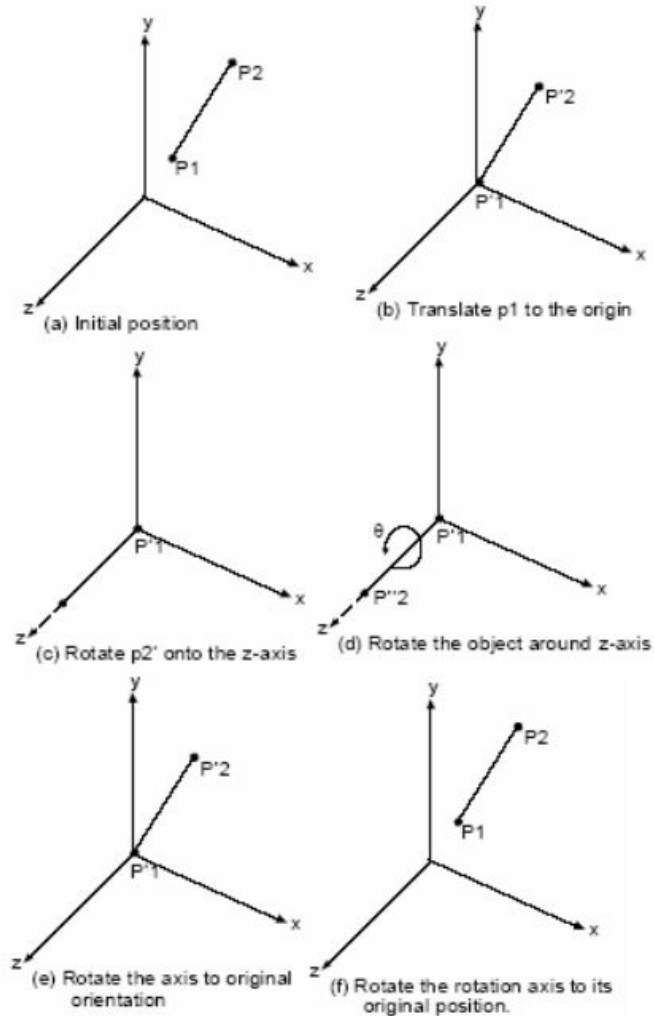
► General 3D rotation (Rotation about any arbitrary axis)

➤ Not parallel to any of the co-axis:

When an object is to be rotated about an axis that is not parallel to one of the co-ordinate axes, we need to perform some series of transformation.

1. Translate the object such that rotation axis passes through co-ordinate origin.
2. Rotate the object such that axis of rotation coincides with one of the co-ordinate axis.
3. Perform the specific rotation about selected coordinate axis.
4. Apply inverse rotation to bring the rotation axis back to its original orientation.
5. Apply inverse translation to bring the rotation axis back to its original position.

► General 3D rotation (Rotation about any arbitrary axis)



► 3D Scaling:

- 3D Scaling

Matrix representation for scaling transformation of a position $P = (x, y, z)$ relative to the coordinate origin can be written as;

$$\begin{bmatrix} x' \\ y' \\ z' \\ w \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix}$$

► 3D Scaling ► Scaling through any fixed position :

→ For scaling of point $P(x, y, z)$ w.r.t to fixed point (x_f, y_f, z_f) can be represented with the following transformation.

1. Translate the fixed point to the origin. $T(-x_f, -y_f, -z_f)$
2. Apply scaling w.r.to origin. $S(s_x, s_y, s_z)$
3. Translate the fixed point back to its original position. $T(x_f, y_f, z_f)$

$$= \begin{bmatrix} 1 & 0 & 0 & x_f \\ 0 & 1 & 0 & y_f \\ 0 & 0 & 1 & z_f \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & -x_f \\ 0 & 1 & 0 & -y_f \\ 0 & 0 & 1 & -z_f \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} s_x & 0 & 0 & (1-s_x)x_f \\ 0 & s_y & 0 & (1-s_y)y_f \\ 0 & 0 & s_z & (1-s_z)z_f \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

► 3D Reflection:

▪ 3D Reflection

In 3D-reflection the reflection takes place about a plane.

(a) About xy-plane (z-axis)

This transformation changes the sign of the z coordinates, leaving the x and y coordinate values unchanged.

$$R_{fxy} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

(b) About xz-plane (y-axis)

This transformation changes the sign of the y coordinates, leaving the x and z coordinate values unchanged.

$$R_{fyz} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

► 3D Reflection:

(c) About yz-plane (x-axis)

This transformation changes the sign of the x coordinates, leaving the y and z coordinate values unchanged.

$$R_{fyz} = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

About any plane:

Step-1: Translate the plane such that it passes through origin i.e. normal vector passes through origin.

Step-2: Rotate the plane such that normal vector lies on one of the co-ordinate axis.

Step-3: Perform reflection about the plane whose normal vector is one of the co-ordinate axis.

Step-4: Rotate back the plane such that normal vector takes its original orientation.

Step-5: Translate back the plane to its original position.

► 3D Shearing:

- 3D Shearing

Shearing transformations are used to modify objects shape.

(a) Z-axis shearing

This transformation alters x and y coordinates values by amount that is proportional to the z values while leaving z coordinate unchanged.

$$x' = x + s_{hx}z$$

$$y' = y + s_{hy}z$$

$$z' = z$$

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & s_{hx} & 0 \\ 0 & 1 & s_{hy} & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

► 3D Shearing:

(b) X-axis shearing

This transformation alters y and z coordinates values by amount that is proportional to the x values while leaving x- coordinate unchanged.

$$x' = x$$

$$y' = y + s_{hy}x$$

$$z' = z + s_{hz}x$$

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ s_{hy} & 1 & 0 & 0 \\ s_{hz} & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

► 3D Shearing:

(c) *Y- axis shearing*

This transformation alters x and z coordinates values by amount that is proportional to the y values while leaving y- coordinate unchanged.

$$x' = x + s_{hx}y$$

$$y' = y$$

$$z' = z + s_{hz}y$$

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & s_{hx} & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & s_{hz} & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

- Q. A homogeneous coordinate point $P(3, 2, 1)$ is translated in x, y, z direction by $-2, -2$ & -2 unit respectively followed by successive rotation of 60° ccw about x - axis. Find the final position of homogeneous coordinate.

Q. A homogeneous coordinate point $P(3, 2, 1)$ is translated in x, y, z direction by $-2, -2$ & -2 unit respectively followed by successive rotation of 60° ccw about x - axis. Find the final position of homogeneous coordinate.

Solution:

Here,

$$t_x = -2$$

$$t_y = -2$$

$$t_z = -2$$

$$T = \begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_x(60^\circ) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos 60 & -\sin 60 & 0 \\ 0 & \sin 60 & \cos 60 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & -\frac{\sqrt{3}}{2} & 0 \\ 0 & \frac{\sqrt{3}}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Q. A homogeneous coordinate point P(3, 2, 1) is translated in x, y, z direction by -2, -2 & -2 unit respectively followed by successive rotation of 60° about x- axis. Find the final position of homogeneous coordinate.

Composite transformation

$$R_x(60^\circ).T = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & -\frac{\sqrt{3}}{2} & 0 \\ 0 & \frac{\sqrt{3}}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & \frac{1}{2} & -\frac{\sqrt{3}}{2} & -1 + \sqrt{3} \\ 0 & \frac{\sqrt{3}}{2} & \frac{1}{2} & -\sqrt{3} - 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Now,

$$P' = M.P = \begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & \frac{1}{2} & -\frac{\sqrt{3}}{2} & -1 + \sqrt{3} \\ 0 & \frac{\sqrt{3}}{2} & \frac{1}{2} & -\sqrt{3} - 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ \frac{\sqrt{3}}{2} \\ -0.5 \\ 1 \end{bmatrix}$$



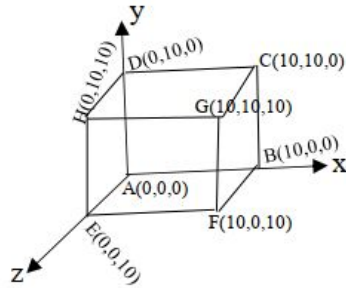
▶ **Q. A cube of length 10 units having one of its corner at origin (0, 0, 0) & three edges along principal axis. If the cube is to be rotated about z-axis by an angle of 45° in counter clockwise direction, calculate the new position of point.**



Q. A cube of length 10 units having one of its corner at origin (0, 0, 0) & three edges along principal axis.

If the cube is to be rotated about z-axis by an angle of 45° in counter clockwise direction, calculate the

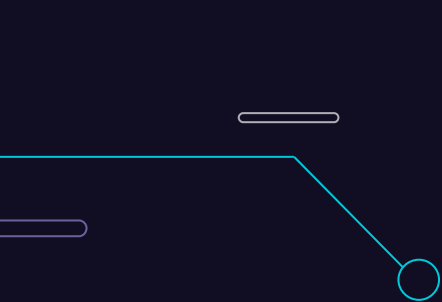
Solution:



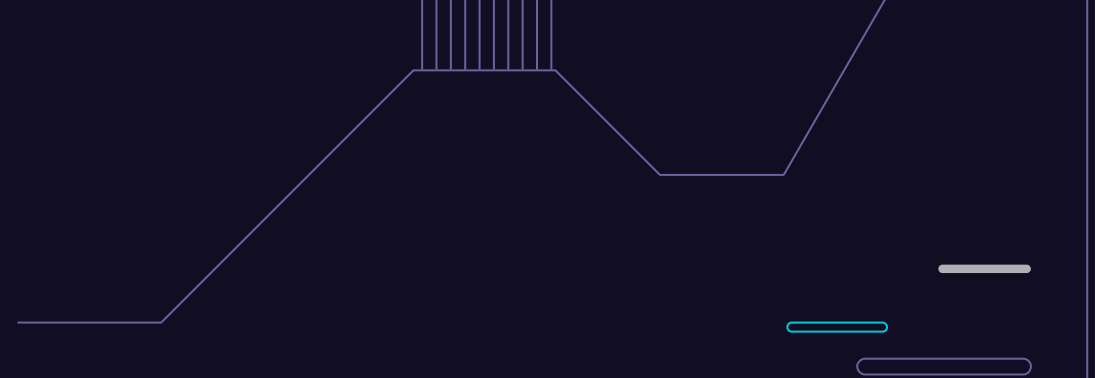
$$R_z(45^\circ) = \begin{bmatrix} \cos 45 & -\sin 45 & 0 & 0 \\ \sin 45 & \cos 45 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.707 & -0.707 & 0 & 0 \\ 0.707 & 0.707 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Now,

$$\begin{bmatrix} 0.707 & -0.707 & 0 & 0 \\ 0.707 & 0.707 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A & B & C & D & E & F & G & H \\ 0 & 10 & 10 & 0 & 0 & 10 & 10 & 0 \\ 0 & 0 & 10 & 10 & 0 & 0 & 10 & 10 \\ 0 & 0 & 0 & 0 & 10 & 10 & 10 & 10 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 7.07 & 0 & -7.07 & 0 & 7.07 & 0 & -7.07 \\ 0 & 7.07 & 14.14 & 7.07 & 0 & 7.07 & 14.14 & 7.07 \\ 0 & 0 & 0 & 0 & 10 & 10 & 10 & 10 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$



▶ 3- Dimensional Viewing



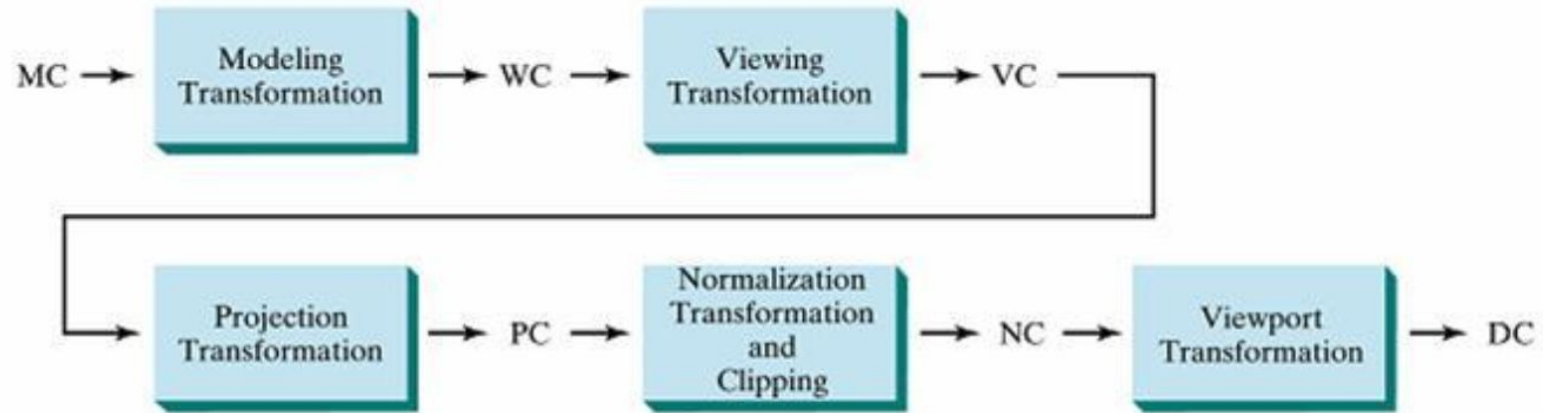
4.3

► 3D Viewing:

Definition: In 3D viewing, we specify a view volume in the world, **a projection onto a projection plane**, and a viewport on the view surface.

- ❖ Conceptually, objects in 3D world are clipped against the 3D view volume and are then projected. The **contents of the projection** of the view volume onto the **projection plane**, called the **window**, are then transformed (mapped) into the viewport for display.
- ❖ Viewing in 3D involves the following considerations:
 - We can view an object from any **spatial position**. E.g. In front of an object, Behind the object, In the middle of a group of the objects, Inside an object.
 - 3D descriptions of objects **must be projected onto the flat viewing surface** of the output device.
 - The clipping boundaries enclose a volume of space.

► 3D Viewing Pipeline:



General three-dimensional transformation pipeline, from modeling coordinates to world coordinates to viewing coordinates to projection coordinates to normalized coordinates and, ultimately, to device coordinates.

► 3D Viewing:

❖ Modeling & Viewing Transformations:

- **3D Transformations:** Used to perform modeling and viewing transformations.
- **Viewing-Coordinate System:** Acts as a reference for specifying the **observer's viewing position** and the **projection plane's position**.

❖ Projection Operations:

- Convert 3D viewing-coordinate descriptions to 2D coordinates on the projection plane.
- **Often combined with:**
 - Clipping
 - Visual-surface identification
 - Surface rendering

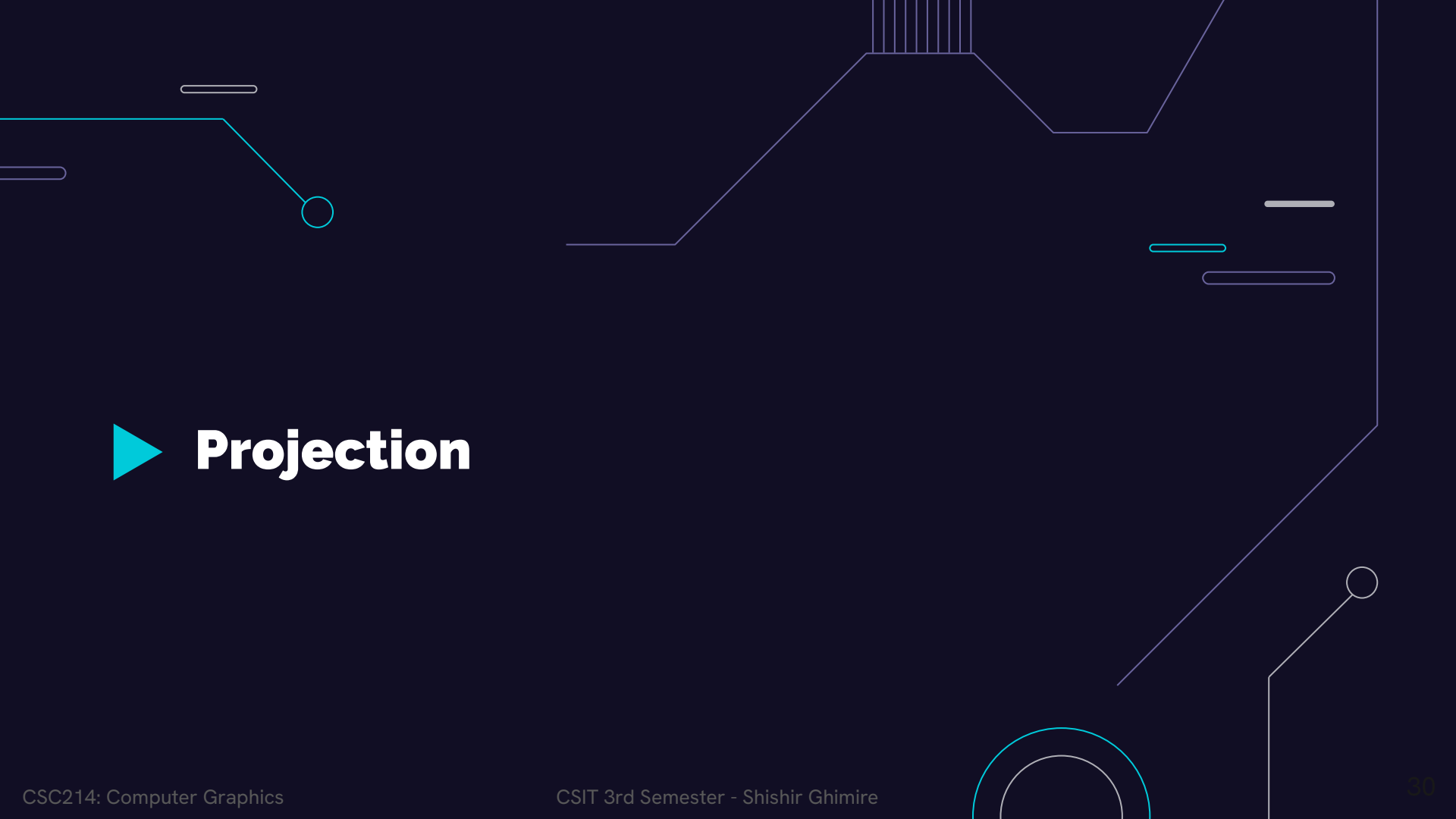
❖ Normalization Transformation & Clipping:

- **Normalization Transformation:** Maps coordinate positions on the projection plane to normalized device coordinates.
- **Clipping & Viewport Transformation:** Maps normalized device coordinates to the output device.

► 3D Viewing:

❖ Window to Screen Viewing Transformation:

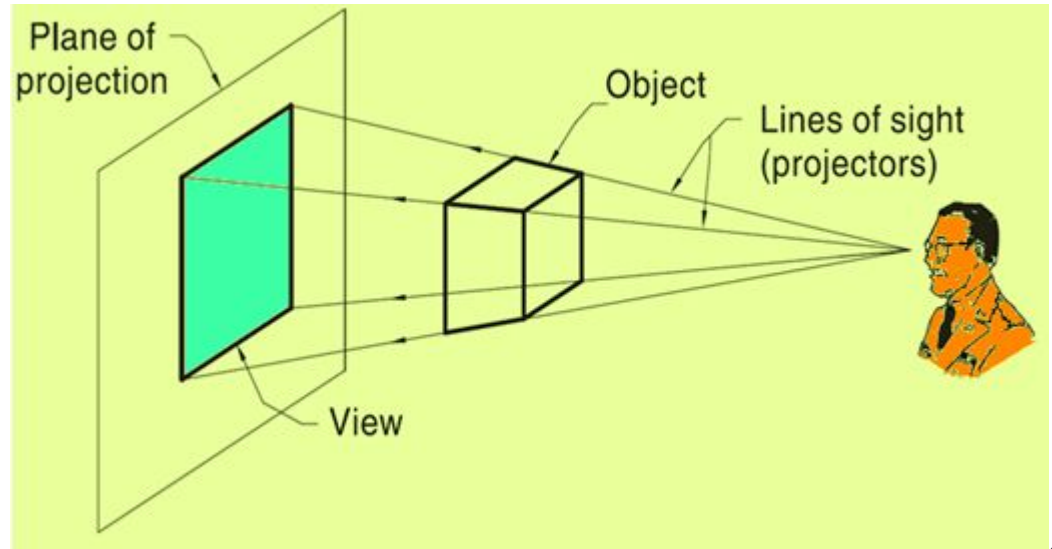
- In 3D Viewing pipeline, the first step after a scene has been constructed is to **transfer object descriptions** to the **viewing- coordinate reference frame**.
- Conversion of object descriptions from world to viewing coordinates is equivalent to following series of transformation:
 - Translate the viewing coordinate origin to the origin of the world coordinate system.
 - Apply rotation to align the x-view, y-view, and z-view with the world x_w , y_w , and z_w -axes respectively.



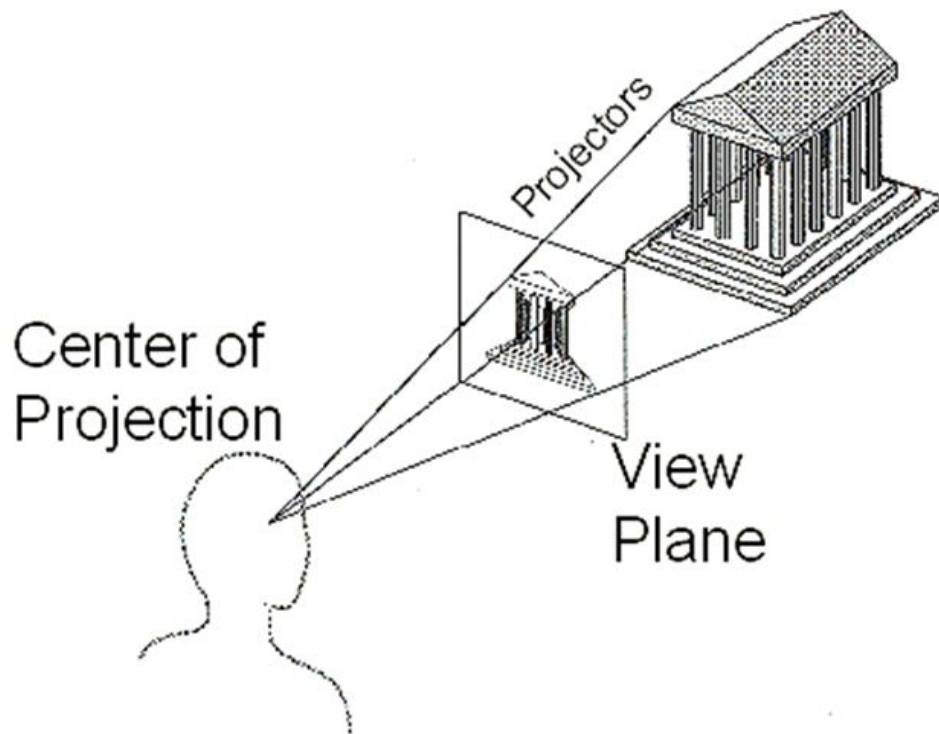
► Projection

► Projection:

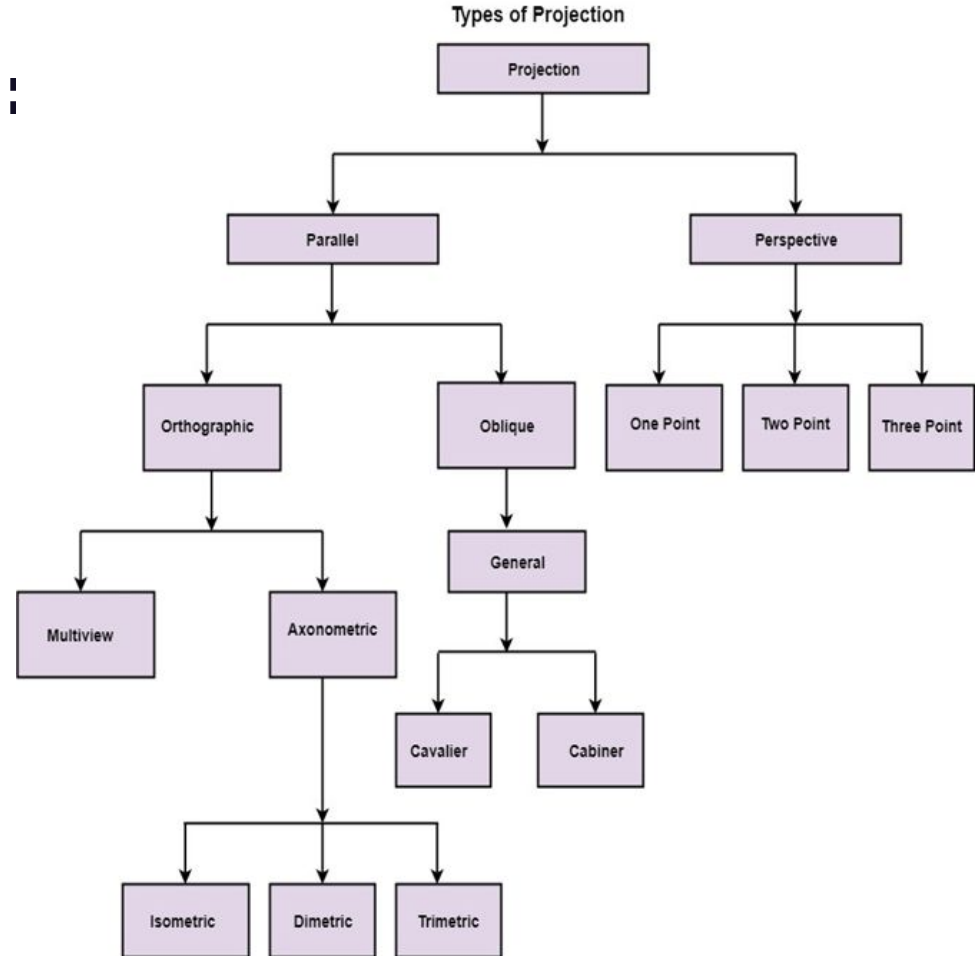
- ❖ Projection is any method of **mapping** three dimensional (3D) objects into two dimensional (2D) view plane (screen).
- ❖ In general, projection transforms a N-dimension points to N-1 dimensions.
- ❖ **Two Types** of Projections:
 1. Parallel Projection
 2. Perspective Projection



► Projection:



► Projection ► Types :



► Projection ► Types :

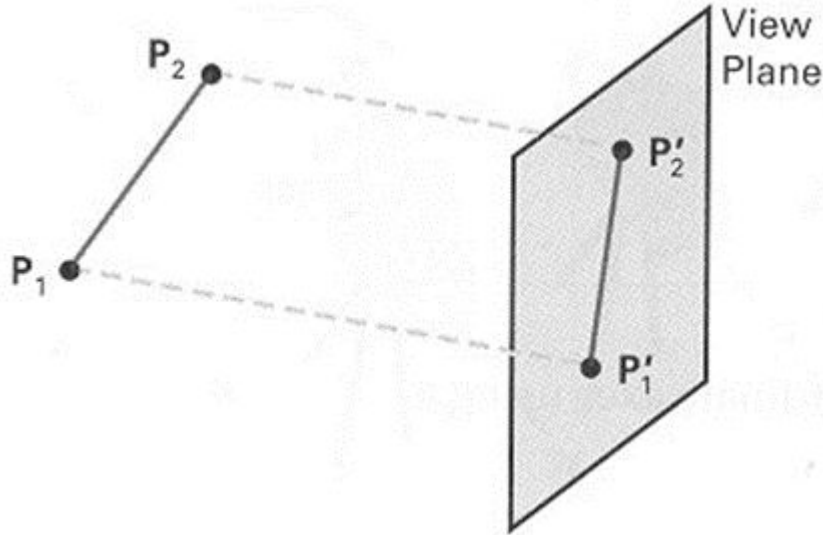


Figure: Parallel Projection

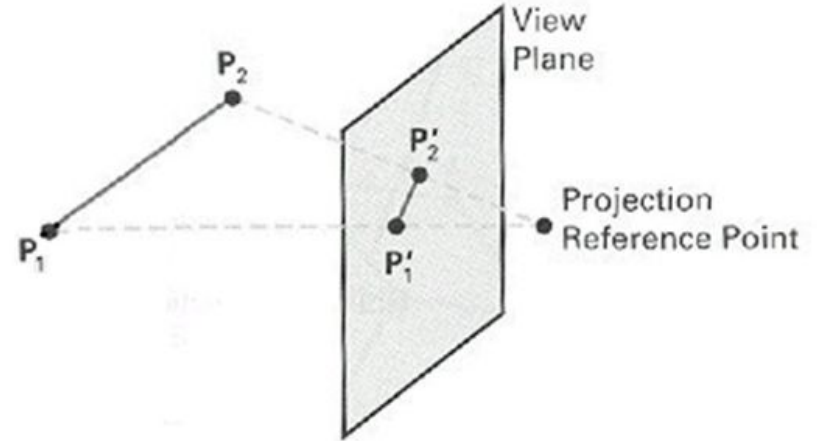


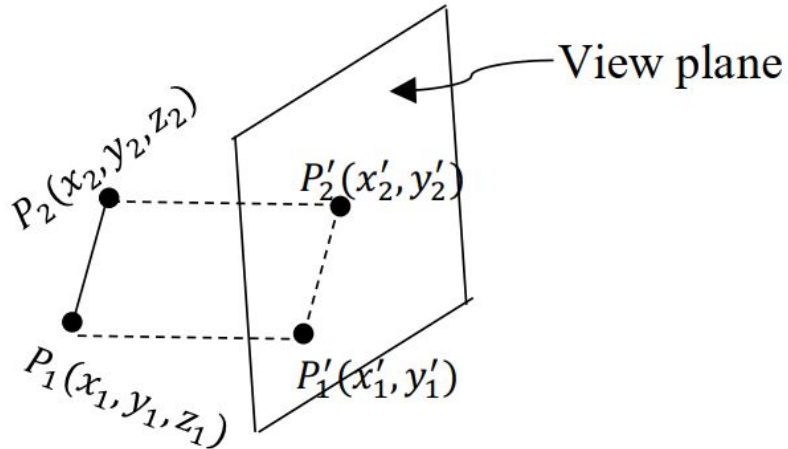
Figure: Perspective Projection

► Projection ► Parallel Projection :

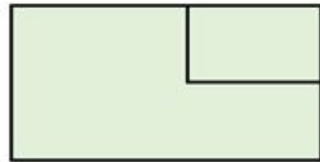
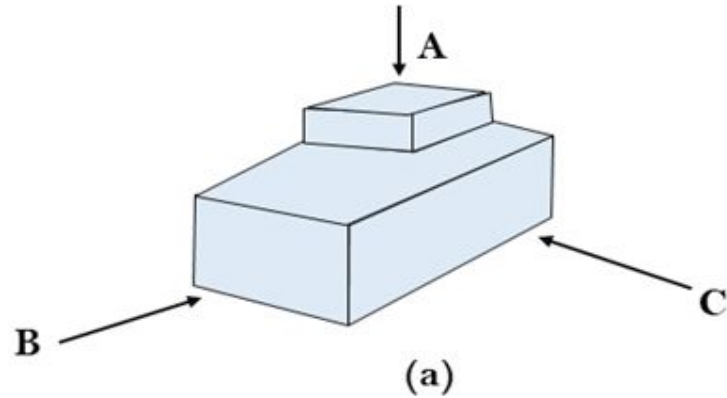
- ❖ In parallel projection, coordinate positions are transformed to view plane **along parallel line**.
 - A parallel projection **preserves relative proportion of objects** so that accurate views of various sides of an object are obtained but doesn't give realistic representation of the 3D objects.
 - Can be used for exact measurement so parallel lines remain parallel.

Figure:

Parallel projection of an object to the view plane.



► Projection ► **Parallel Projection** :



Parallel projection from
top i.e. A direction

(b)



Parallel projection from
top i.e. B direction

(c)



Parallel projection from
top i.e. C direction

(d)

► **Projection** ► **Parallel Projection** :

❖ On the basis of angle made by projection line with view plane, there are two types of parallel projection.

1. Orthographic Parallel Projection
2. Oblique Parallel Projection

► Projection ► Parallel Projection ► **Types :**

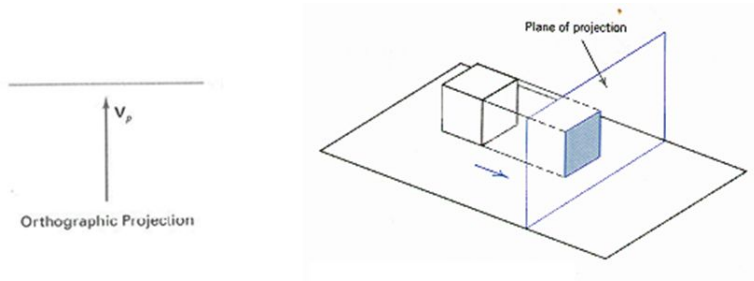


Figure: Orthographic Projection

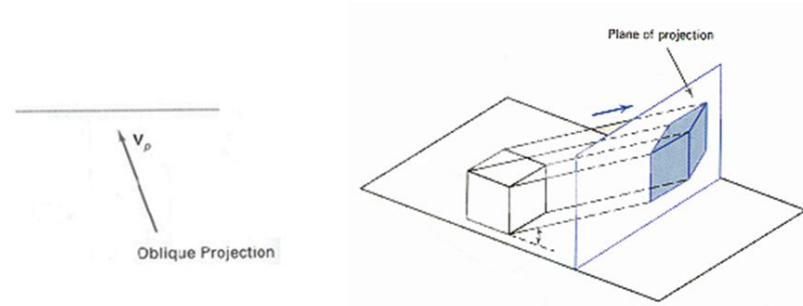
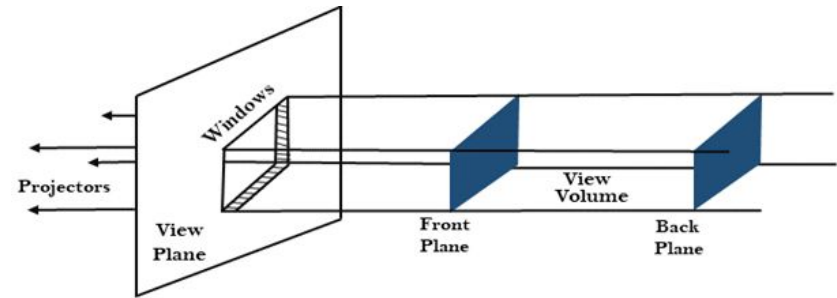
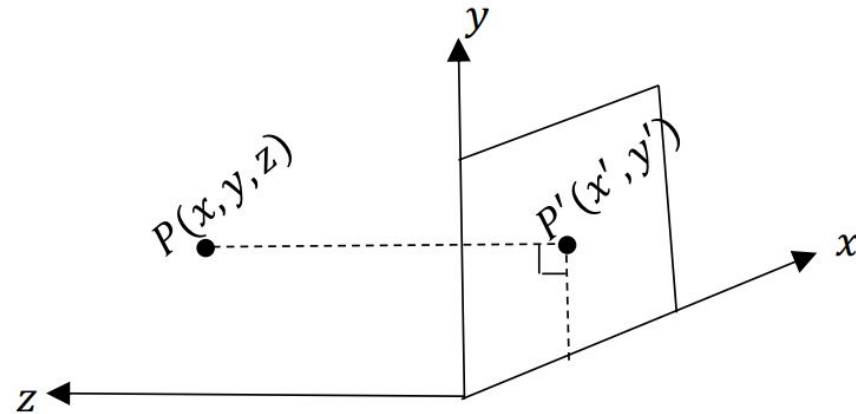


Figure: Oblique Projection

► Parallel Projection ► Orthographic Projection :

- ❖ When the projection lines are **perpendicular to view plane**, the projection is orthographic parallel projection.



(a) Viewing Volume in orthographic projection

► **Parallel Projection** ► **Orthographic Projection** :

Here, after projection of $P(x, y, z)$ on XY-plane we get $P'(x', y')$ where,

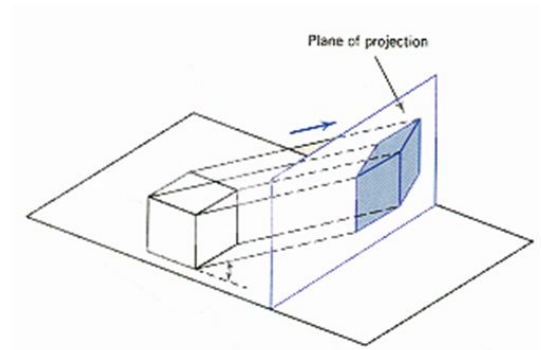
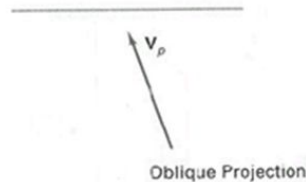
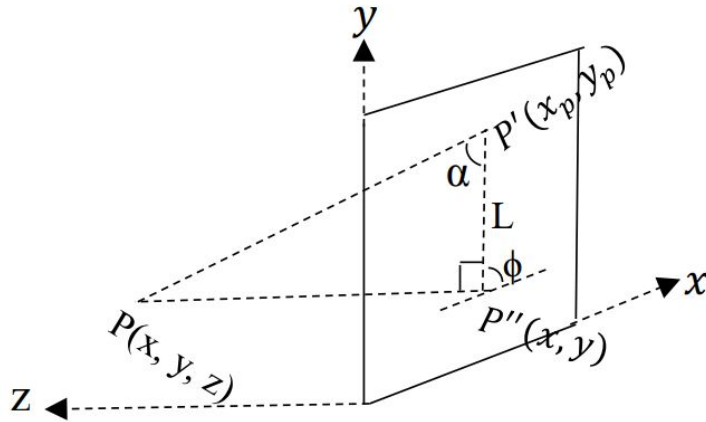
$$x' = x, y' = y \text{ \& } z=0$$

In homogenous coordinate form,

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

► Parallel Projection ► Oblique Projection :

- ❖ Here, projection lines make **certain angle to view plane** (non perpendicular i.e curve surface).
- ❖ Oblique projection is specified with two angle ' α ' & ' ϕ ' where ' α ' is the angle made by **projection line with view plane line** (L) which is formed by joining oblique projected point $P'(x_p, y_p)$ and orthogonal projected point $P''(x, y)$ on point (x, y) on view plane & ' ϕ ' is the angle between '**L**' & **horizontal direction (x-axis)** of view plane.



► Parallel Projection ► Oblique Projection :

The projected point $P'(x_p, y_p)$ on XY plane is given by,

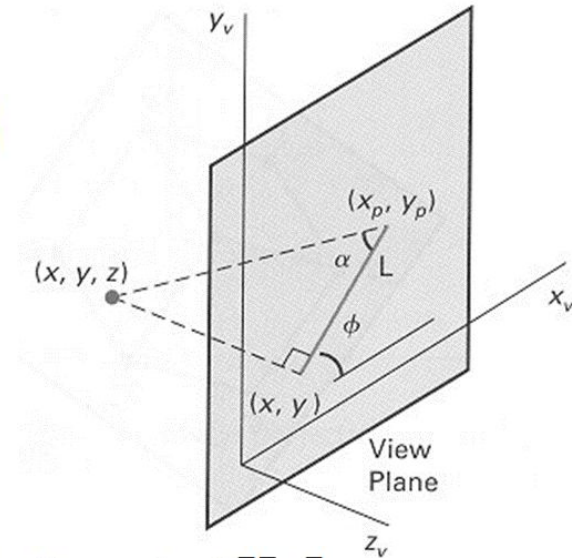
$$x_p = x + L \cos \phi, y_p = y + L \sin \phi$$

$$\tan \alpha = \frac{z}{L}, L = \frac{z}{\tan \alpha} = zL_1$$

$$x_p = x + z(L_1 \cos \phi)$$

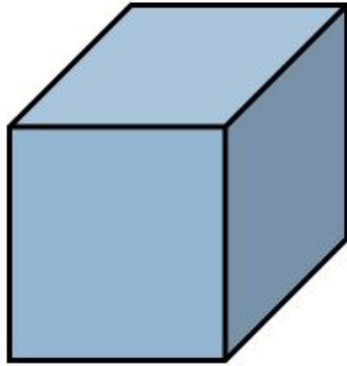
$$y_p = y + z(L_1 \sin \phi)$$

Where L_1 is the inverse of $\tan \alpha$, which is also the value of L when $z=1$



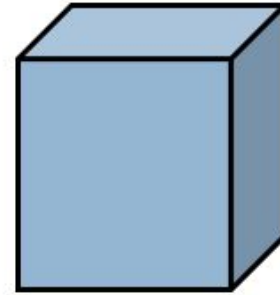
$$\begin{bmatrix} x_p \\ y_p \\ z_p \\ w_p \end{bmatrix} = \begin{bmatrix} 1 & 0 & L_1 \cos \phi & 0 \\ 0 & 1 & L_1 \sin \phi & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

► Parallel Projection ► Oblique Projection ► Types :



Cavalier

(DOP at 45° , $\tan\alpha=1$)

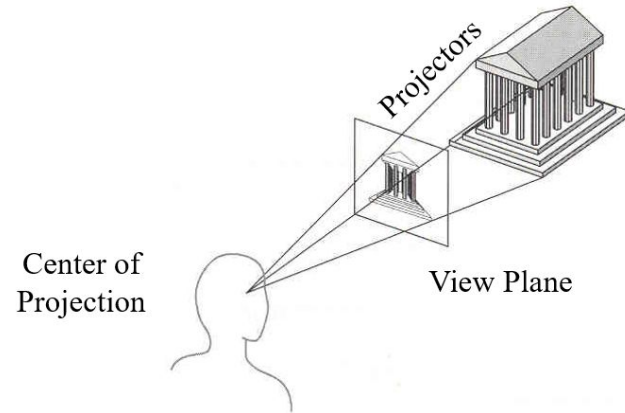
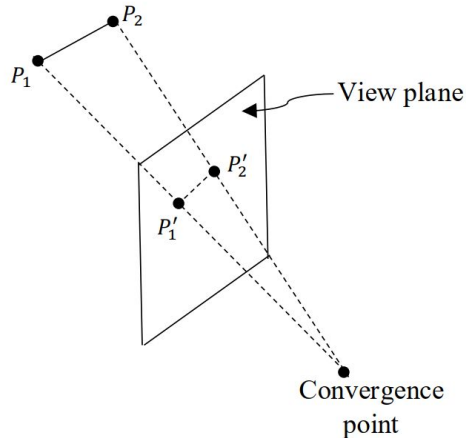


Cabinet

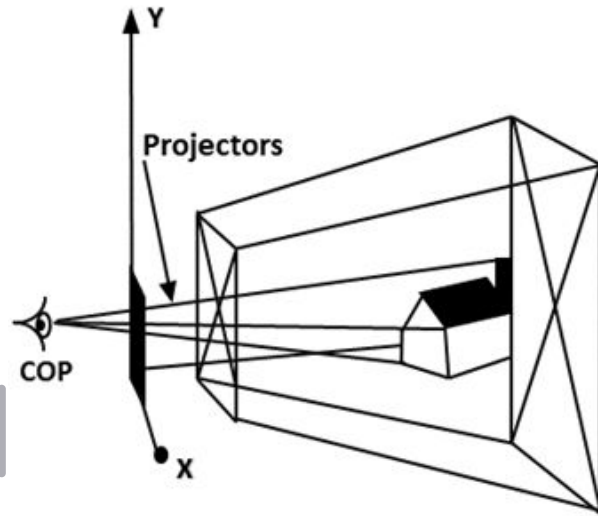
(DOP at 63.4° , $\tan\alpha=2$)

► Projection ► Perspective Projection :

- ❖ In perspective projection, objects positions are **transformed to the view plane** along lines that **converge to point behind view plane**.
- A perspective projection **produces realistic views** but **does not preserve** relative proportions. Projections of distance objects, from view plane are **smaller** than the projections of objects of the same size that are closer to the projection place.

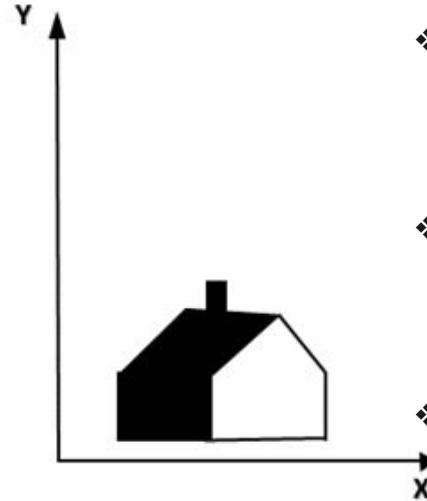


► Projection ► Perspective Projection :



View Volume

(a)



Viewing plane

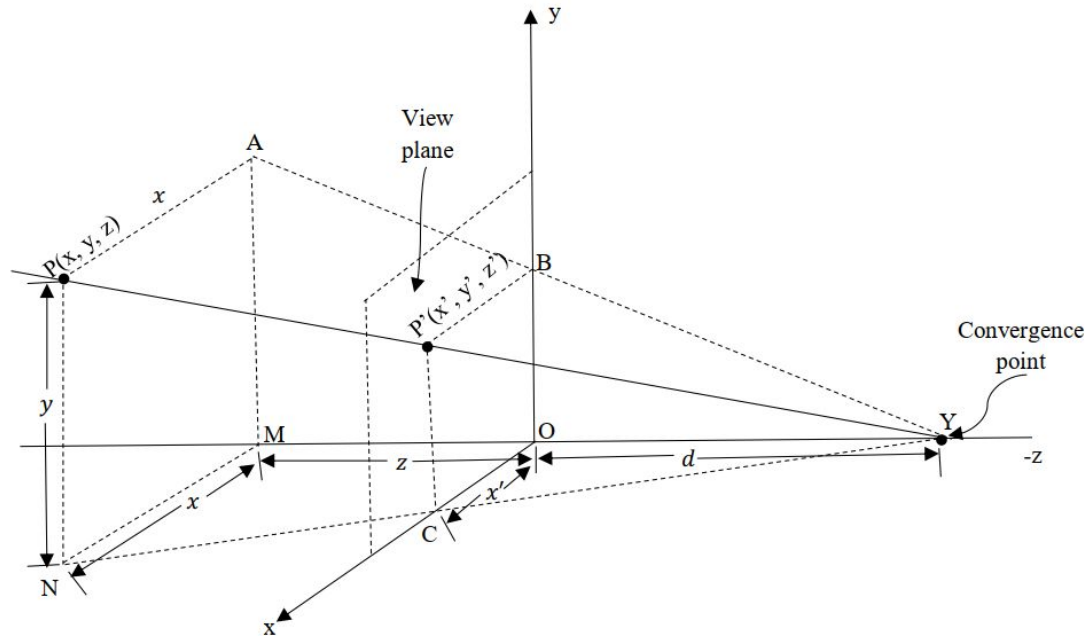
(b)

Fig: Perspective Projection

- ❖ **View plane:** It is an area of world coordinate system which is projected into viewing plane.
- ❖ **Center of Projection:** It is the location of the eye on which projected light rays converge.
- ❖ **Projectors:** It is also called a projection vector. These are rays start from the object scene and are used to create an image of the object on viewing or view plane.

► Projection ► Perspective Projection :

- ❖ Let $P(x, y, z)$ be projected on X-Y view plane by perspective projection at $P'(x', y', z')$ & projected line converges point 'y' and z-axis at is distance 'd' from XY view plane.



► Projection ► Perspective Projection :

We have to find the value of $P'(x', y', z')$ from similar triangle NMY & COY.

$$\frac{NM}{MY} = \frac{CO}{OY}$$

$$\frac{x}{z+d} = \frac{x'}{d}$$

$$\therefore x' = x \frac{d}{z+d}$$

Similarly, from similar triangle AMY & BOY

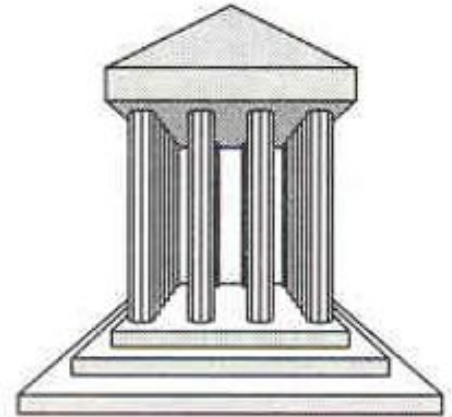
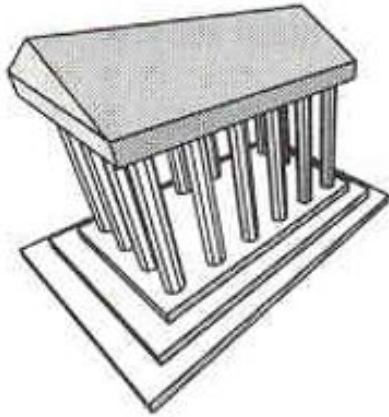
$$\frac{AM}{MY} = \frac{BO}{BY}$$

$$\frac{y}{z+d} = \frac{y'}{d}$$

$$\therefore y' = y \frac{d}{z+d}$$

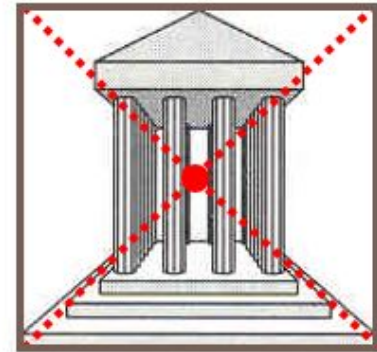
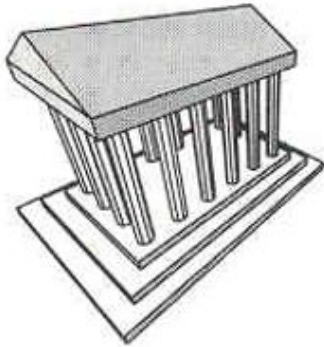
► Projection ► **Perspective Projection** :

❖ How many vanishing point?



► Projection ► **Perspective Projection** :

❖ How many vanishing point?

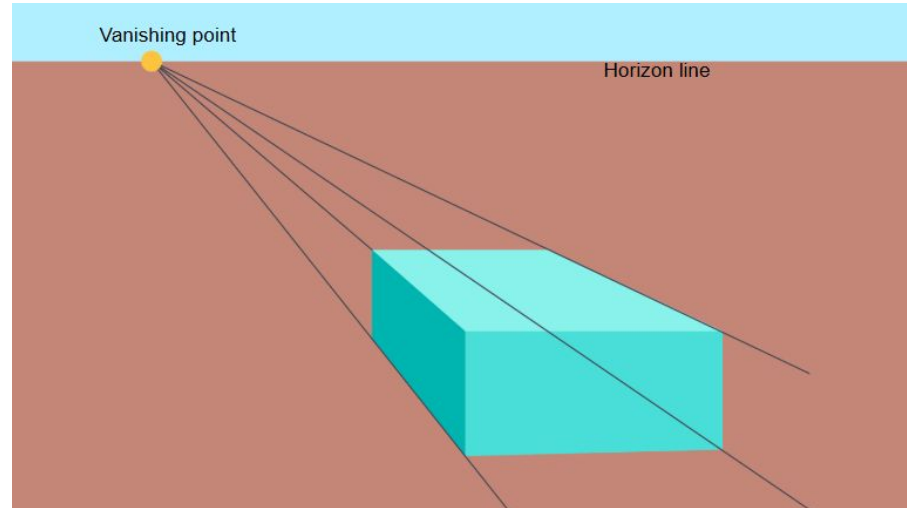


One-point
perspective

► Projection ► Perspective Projection :

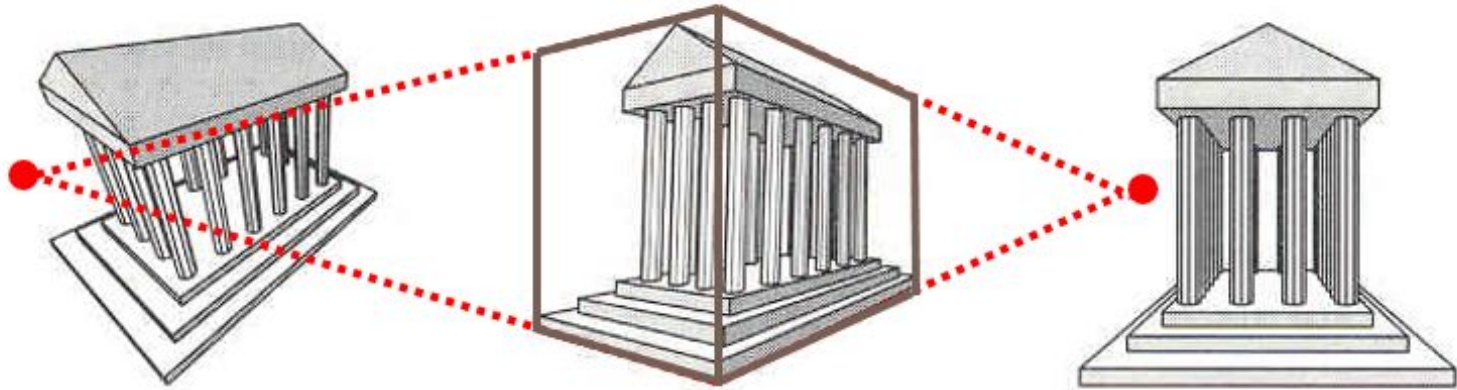
❖ One-point perspective

- In this projection, all lines and objects in the scene converge towards a **single vanishing point** on the horizon line, providing a sense of depth. For example, the lines converge to a yellow point in the following image.



► Projection ► **Perspective Projection** :

❖ How many vanishing point?

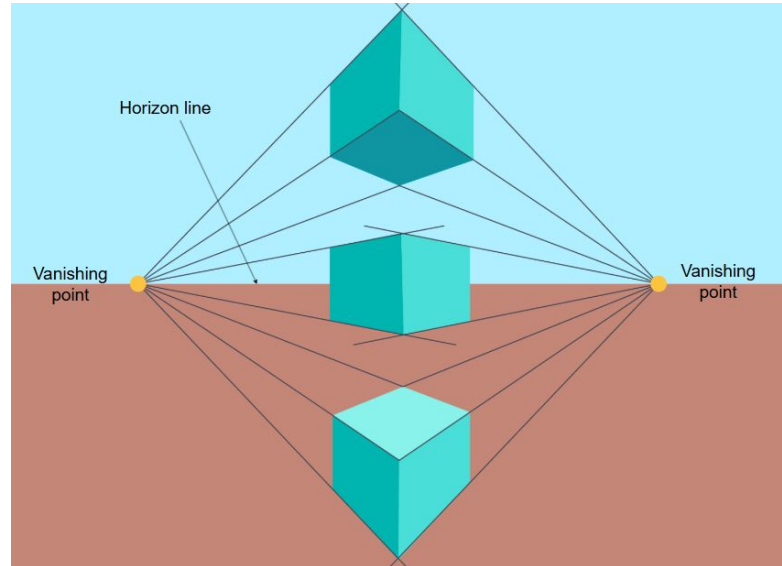


Two-point
perspective

► Projection ► **Perspective Projection** :

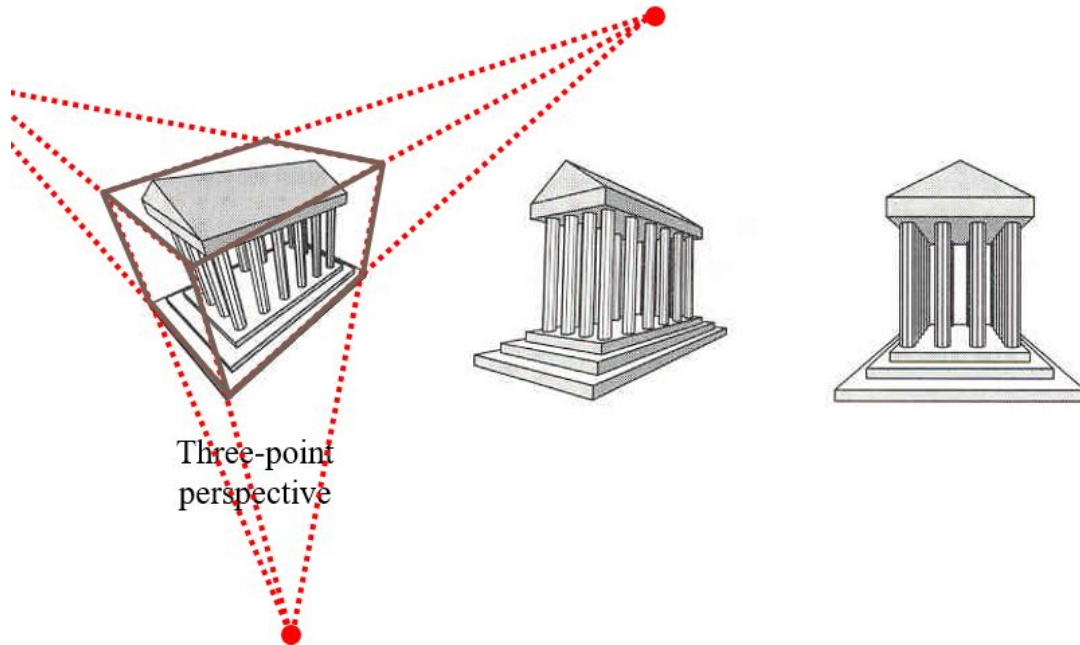
◆ Two-point perspective

- In the two-point perspective, the receding lines converge to two vanishing points on the horizon line. For example, the lines converge to two yellow points in the following image:



► Projection ► Perspective Projection :

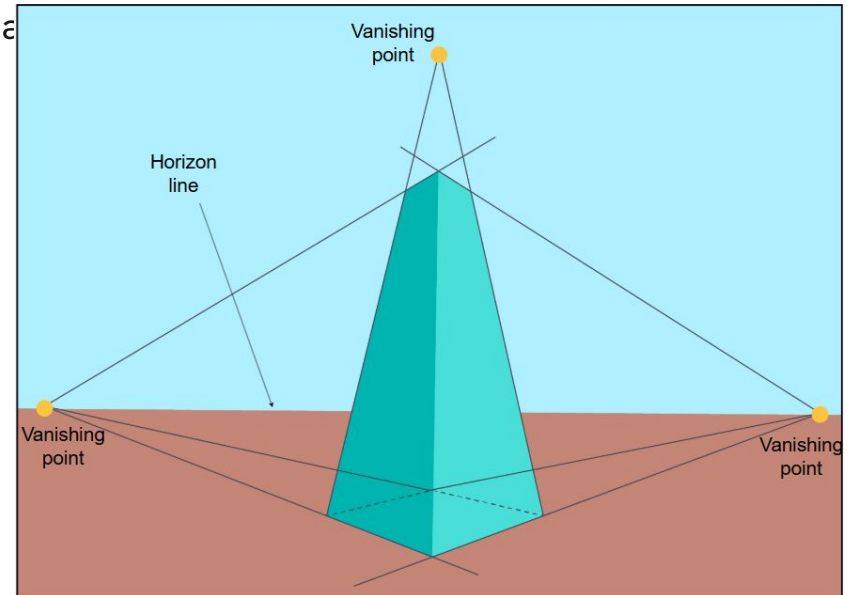
❖ How many vanishing point?



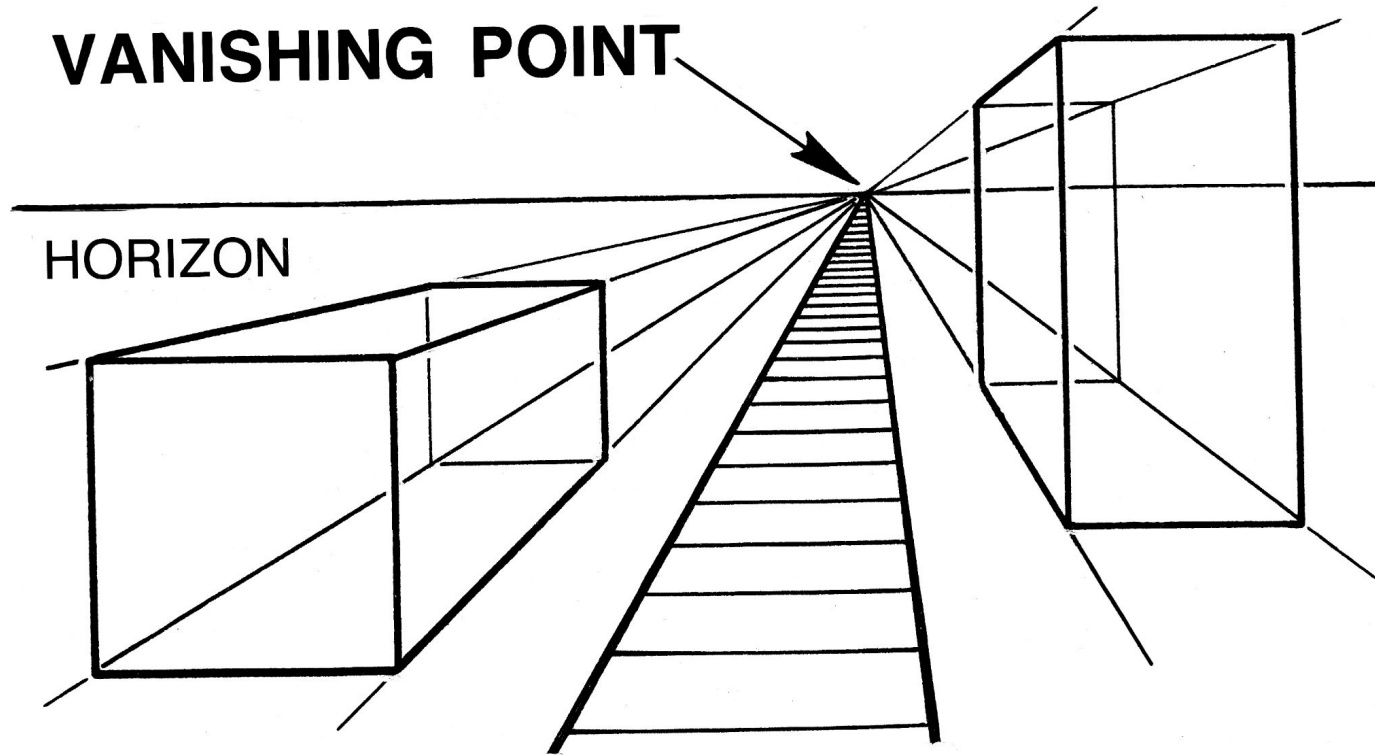
► Projection ► Perspective Projection :

◆ Three-point perspective

- In the three-point perspective, the receding lines converge to three vanishing points, two on the horizon line and one above or below it. For example, the lines converge to three yellow points in the following image

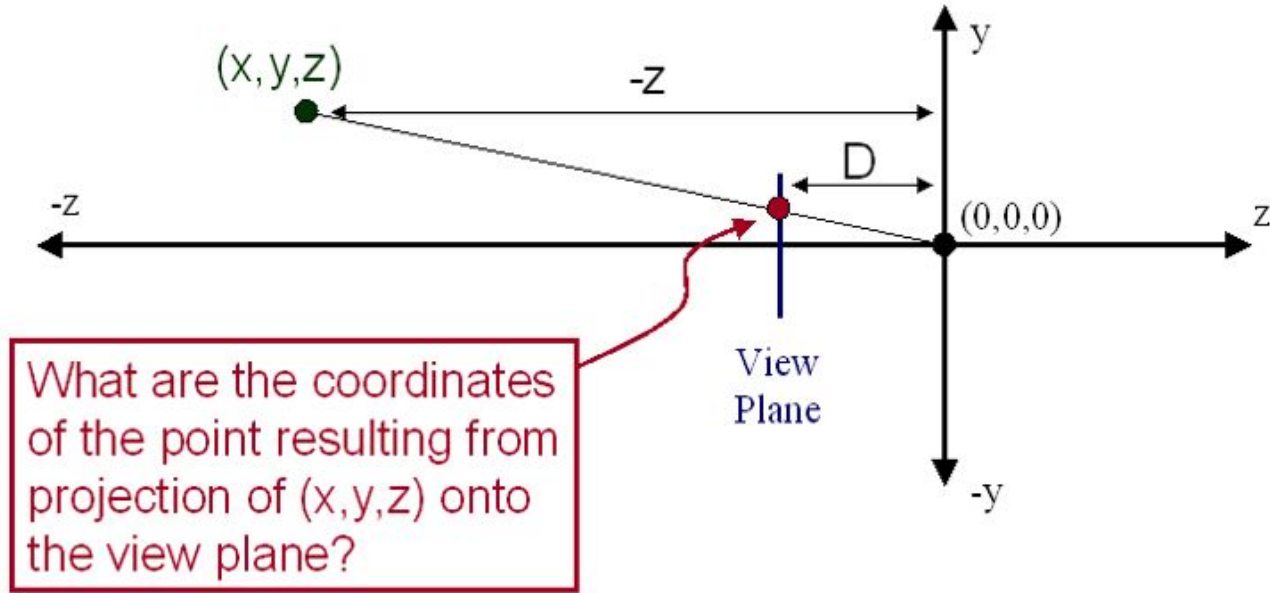


► Projection ► **Perspective Projection** :



► Projection ► Perspective Projection :

- ❖ Compute 2D coordinates from 3D coordinates with similar triangles



► Projection ► Perspective Projection :

- ❖ Compute 2D coordinates from 3D coordinates with similar triangles

Look at the ray from O to P . In the xz -plane:

- Triangle $\triangle O(0,0), (x,0,z), (0,0,z)$
- Triangle $\triangle O(0,0), (x',0,D), (0,0,D)$

These triangles are **similar** (same angles), so the x -to- z ratios match:

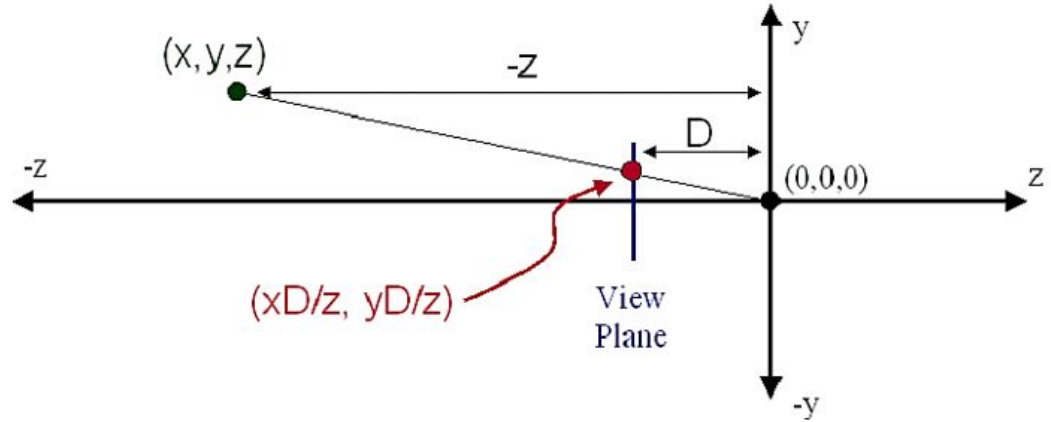
$$\frac{x'}{D} = \frac{x}{z} \Rightarrow x' = \frac{D}{z} x$$

Likewise in the yz -plane:

$$\frac{y'}{D} = \frac{y}{z} \Rightarrow y' = \frac{D}{z} y$$

Result:

$$x' = \frac{D}{z} x, \quad y' = \frac{D}{z} y$$



► Projection ► **Perspective Projection** :

□ 4x4 matrix representation?

$$x_p = xD / z$$

$$y_p = yD / z$$

$$z_p = D$$

$$w_p = 1$$

$$x' = x$$

$$y' = y$$

$$z' = z$$

$$w_p = z/D$$

$$\begin{bmatrix} x_p \\ y_p \\ z_p \\ w_p \end{bmatrix} = \begin{bmatrix} ? & ? & ? & ? \\ ? & ? & ? & ? \\ ? & ? & ? & ? \\ ? & ? & ? & ? \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

► Projection ► **Perspective Projection** :

□ 4x4 matrix representation?

$$x_p = xD / z$$

$$y_p = yD / z$$

$$z_p = D$$

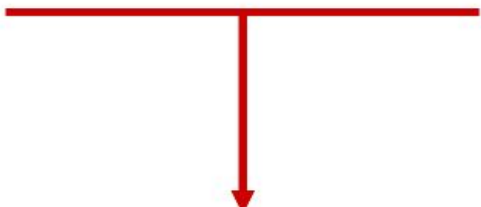
$$w_p = 1$$

$$x' = x$$

$$y' = y$$

$$z' = z$$

$$w_p = z / D$$


$$\begin{bmatrix} x_p \\ y_p \\ z_p \\ w_p \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1/D & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

► Projection ► Perspective Projection :

$$\mathbf{M}_{PER} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{D} & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & \frac{1}{D} & 1 \end{bmatrix}$$

Perspective
projection

Orthographic
projection

Perspective
transformation

Center of Projection on the x axis

$$\mathbf{M}_{PER} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \frac{1}{D_x} & 0 & 0 & 1 \end{bmatrix}$$

Center of Projection on the y axis

$$\mathbf{M}_{PER} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & \frac{1}{D_y} & 0 & 1 \end{bmatrix}$$

► Projection ► Perspective Projection :

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ r & s & 0 & 1 \end{bmatrix}$$

or

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & s & t & 1 \end{bmatrix}$$

2-point perspectives

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ r & s & t & 1 \end{bmatrix}$$

3-point perspectives

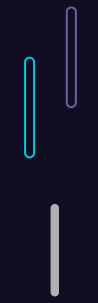
► Parallel vs **Perspective** Projection :

Parallel Projection	Perspective Projection
Coordinate position of object are transferred into view plane along parallel line.	Coordinate positions are transferred into view plane along lines that converges to a point called convergence point.
Relative proportion of object are maintained.	Relative position not maintained.
It gives accurate view of object.	If view plane is nearest to object image appear larger & if view plane is farther image appear smaller.
It doesn't give realistic view of object.	It gives more realistic view of object as perceived by human eye.
Used in engineering and architectural drawing.	Used in building design, rail track design, whose realistic view is needed.

► Assignment -4:

Due Date: 2025 February 9th

- ❖ Derive the relation for three- dimensional translation and rotation.
- ❖ Explain 3D Scaling, 3D Mirroring with practical examples.
- ❖ Explain 3D Viewing pipeline with equation and practical applications.
- ❖ Write homogeneous coordinates for all 3D Transformation.
- ❖ What is projection? List its types.
- ❖ Differentiate between oblique and orthographic projection.
- ❖ Differentiate between parallel and perspective projection with suitable diagram. Illustrate the window to viewport transformation with example. 2080
- ❖ What are the transformation matrix for perspective and parallel projection.



▶ **THANKS!**

Do you have any questions?



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